La Scuola d'Italia / Dr. Huson / IB Math 12 HL
10 November 2026

Nickname:

5.5 Classwork: Proof, Differential Equations and Series (no calculator)

Show all working and give exact answers. If you need more space, continue on lined paper and label the problem and part.

1. [Maximum mark: 13]

   In this question, use these facts about integers: the sum, difference and product of integers are integers; an integer is *even* if it equals $2m$ for some $m \in \mathbb{Z}$, and *odd* if it equals $2m + 1$. No integer is both even and odd.

   (a) Show that $n^3 - n$ is even for every $n \in \mathbb{Z}$. [2]

   (b) Prove by contradiction that the equation $x^3 + x + 1 = 0$ has no integer roots. Begin with a precise statement of what you assume. [5]

   (c) Prove by contradiction that $p^2 - 8q - 3 \neq 0$ for all $p, q \in \mathbb{Z}$. [6]

2. [Maximum mark: 17]

(a) Find $\int \left(x^2+1\right) e^x \, \mathrm{d}x$. [4]

Consider the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} = x^2 + 1 - y$, where $y = 1$ when $x = 0$.

(b) By solving the differential equation, show that $y = x^2 - 2x + 3 - 2e^{-x}$. [5]

(c) Without using the solution from part (b), use the differential equation to find the Maclaurin series for y, up to and including the term in x^3. [5]

(d) Show that the curve $y = x^2 - 2x + 3 - 2e^{-x}$ has a point of inflection at $(0,\ 1)$. [3]

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3. [Maximum mark: 15]

Let $f(x) = \ln(1+x)$, for $x > -1$.

(a) Prove by mathematical induction that

$$f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n} \quad \text{for all } n \in \mathbb{Z}^+.$$

[7]

(b) Hence show that the Maclaurin series for $f(x)$ is [3]

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + \frac{(-1)^{n-1}x^n}{n} + \cdots$$

(c) Find the Maclaurin series for $\ln\left(\frac{1+x}{1-x}\right)$, up to and including the term in x^5. [3]

(d) Hence find $\lim_{x\to 0} \frac{\ln\left(\frac{1+x}{1-x}\right) - 2x}{x^3}$. [2]

4. [Maximum mark: 14]

(a) Let $w = a + bi$, $a, b \in \mathbb{R}$. Prove that $w + w^* = 2\,\mathrm{Re}(w)$ and $ww^* = |w|^2$. [3]

(b) Let $w = re^{i\theta}$. Use de Moivre's theorem to show that $(w^*)^n = (w^n)^*$, $n \in \mathbb{Z}^+$. [3]

(c) Hence prove that $(1+i)^n + (1-i)^n$ is real for all $n \in \mathbb{Z}^+$. [2]

(d) Show that $(1+i)^n + (1-i)^n = 2^{\frac{n}{2}+1} \cos\dfrac{n\pi}{4}$. [4]

(e) Hence find the value of $(1+i)^{12} + (1-i)^{12}$. [2]